pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

More Visualization

WEEK 3 · DAY 2

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Yesterday drew the book. Today finishes the charts.

Yesterday ended with seven kinds of chart

Each one was made with fig, ax = plt.subplots(), drawn on ax, and labelled with units. Two numbers from those charts come back today: the median OAS of the book, and the correlation of duration and OAS.

135.5
bp, the median OAS

-0.11
duration and OAS

1
new library: plotly

A fitted line is a summary of this file. It is not a forecast.

The portfolio manager has three more requests

A line through the scatter, with its slope

- Regression plot
- Joint plot

The bonds behind the boxes and the bars

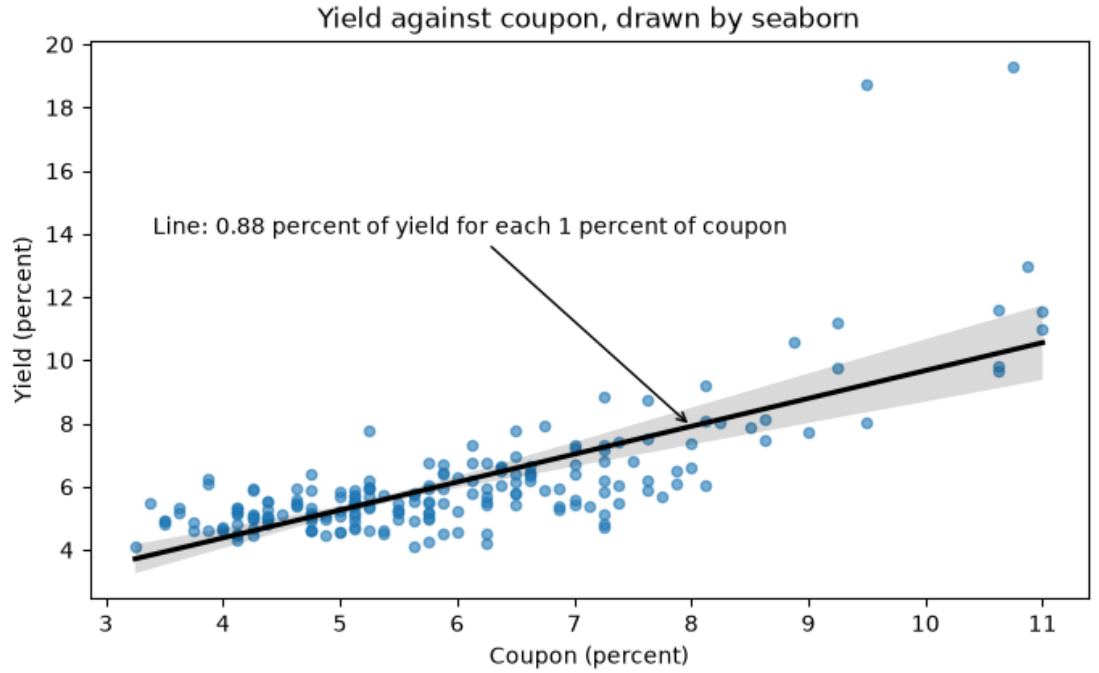
- Violin plot
- Strip and swarm
- Catplot

Charts that are fit for a meeting

- Interactive
- 3D charts
- Customizing

`matplotlib`	and seaborn, as yesterday
`plotly`	new today: charts that respond to the mouse

Regression plot: a line through the scatter



The number	Value	Excel: coupon in G, yield in L
Slope	0.88	`=SLOPE(L2:L201, G2:G201)`
Intercept	0.85	`=INTERCEPT(L2:L201, G2:G201)`
R squared	0.55	`=RSQ(L2:L201, G2:G201)`

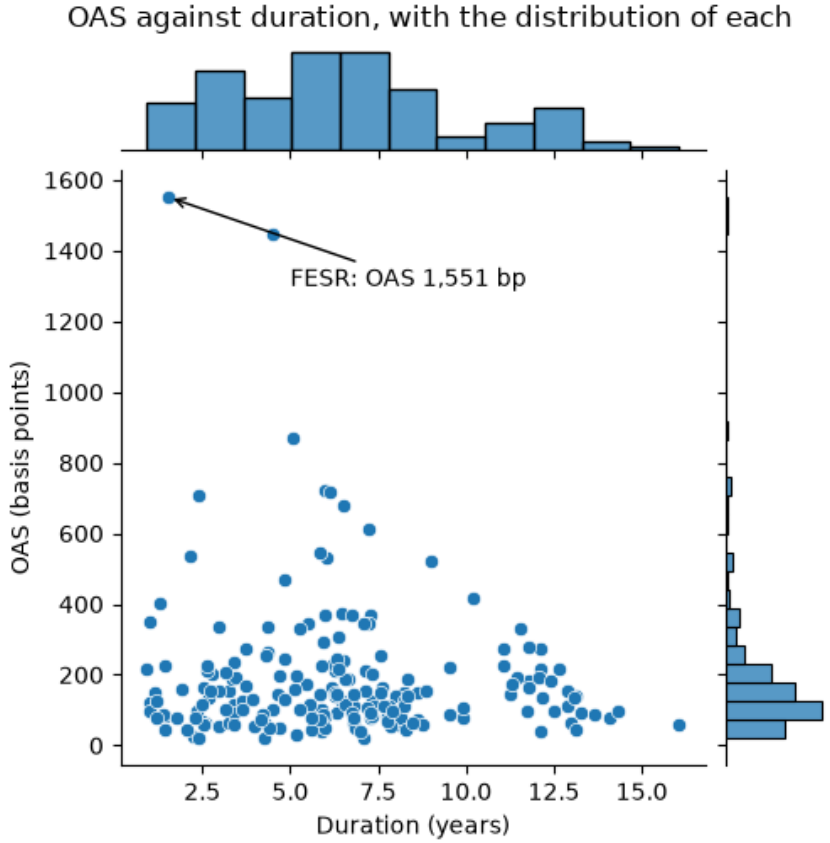
Units: the slope is percent of yield per 1 percent of coupon. The intercept is the line at a coupon of zero, where no bond is. R squared: the line accounts for 55 percent of the variation in yield. The band: at each coupon, the middle 95 percent of 1,000 lines refitted to random redraws. Excel takes the y range first.

In Excel: Chart Elements, Trendline, More Options, Linear

```
slope, intercept = np.polyfit(x, y, 1)   # x coupon, y yield
sns.regplot(data=book, x='coupon', y='yield_pct', ax=ax)
```

The line describes this invented file. It is not a forecast.

Joint plot: two columns, and each on its own




```
grid = sns.jointplot(data=book, x='duration', y='oas',
                     height=5)
```

What to read off it

169

of 200 bonds at 10 years or less

164

of 200 bonds below 250 bp

3

axes: a scatter and two histograms

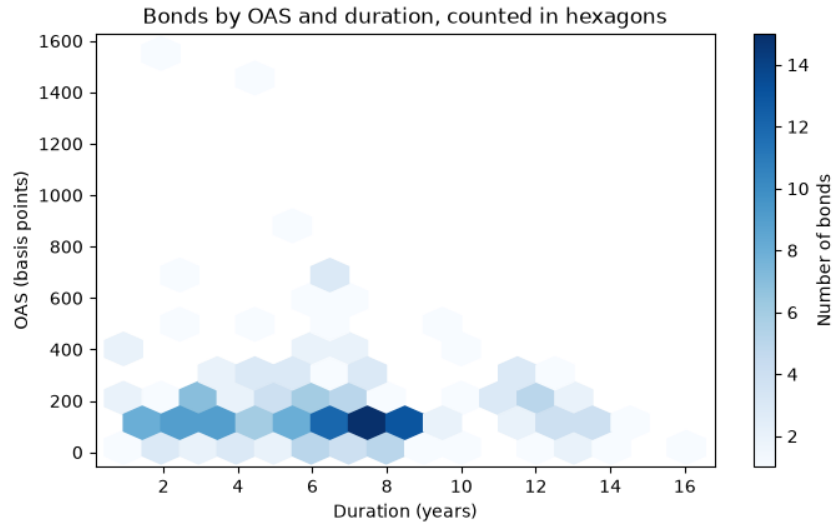
The top histogram belongs to the x axis and the side one to the y axis. The points are dense along the bottom because the OAS column is dense there, at every duration.

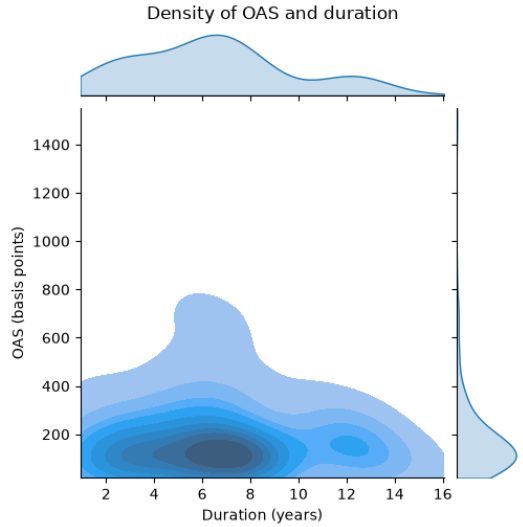
The same in Excel

No single chart. The same picture is three charts on a sheet: a Scatter, a Histogram of duration above it, and a Histogram of OAS beside it.

`=COUNTIF(M2:M201, "<250")` returns 164 `=COUNTIF(N2:N201, "<=10")` returns 169

Crowded points: hexagons and density



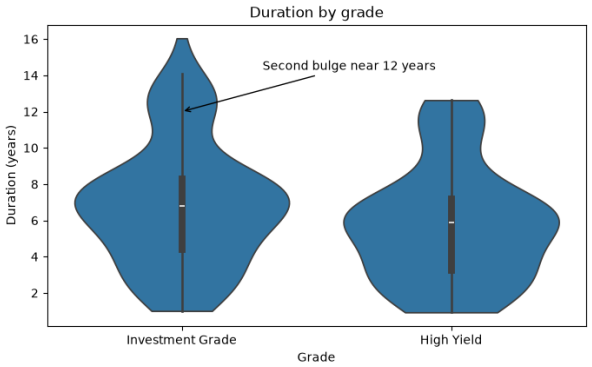



```
hexes = ax.hexbin(book['duration'], book['oas'],
                  gridsize=15, mincnt=1)
fig.colorbar(hexes, ax=ax, label='Number of bonds')
```

Hexagons count the bonds in each part of the plane: the darkest, near 6 to 8 years, hold 15. The density is the outline: a dense area near 6 to 7 years, a smaller one near 12. It is a summary, and the widest bonds barely show.

Violin, strip, and swarm: inside the box

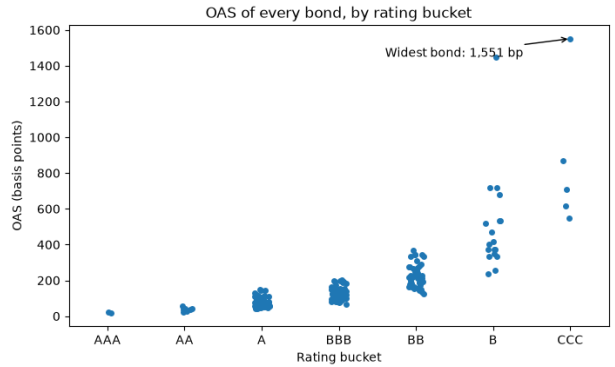
Violin plot: the shape of a group




```
sns.violinplot(data=book, x='grade',
    y='duration', cut=0, ax=ax)
```

A box plot with a mirrored density curve around it. Medians: 6.79 years for Investment Grade, 5.87 for High Yield. The width is not a count. Excel has no violin chart type. The nearest is Box and Whisker.

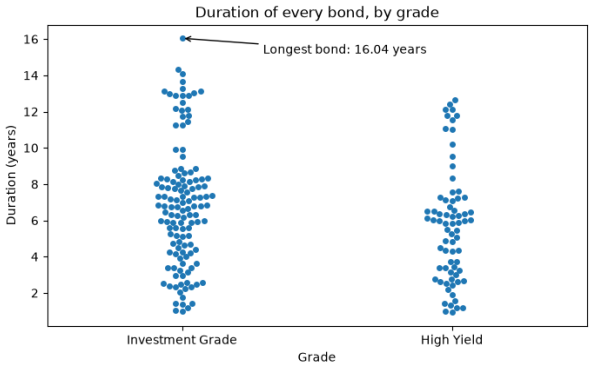
Strip plot: every bond, jittered




```
sns.stripplot(data=book, x='bucket',
    y='oas', order=bucket_order, ax=ax)
```

One point per bond: 2 in AAA, 62 in BBB, 5 in CCC. The two widest, 1,446 and 1,551 bp, stand alone. Sideways position is random. In Excel, =COUNTIF(U2:U201, "CCC") returns 5.

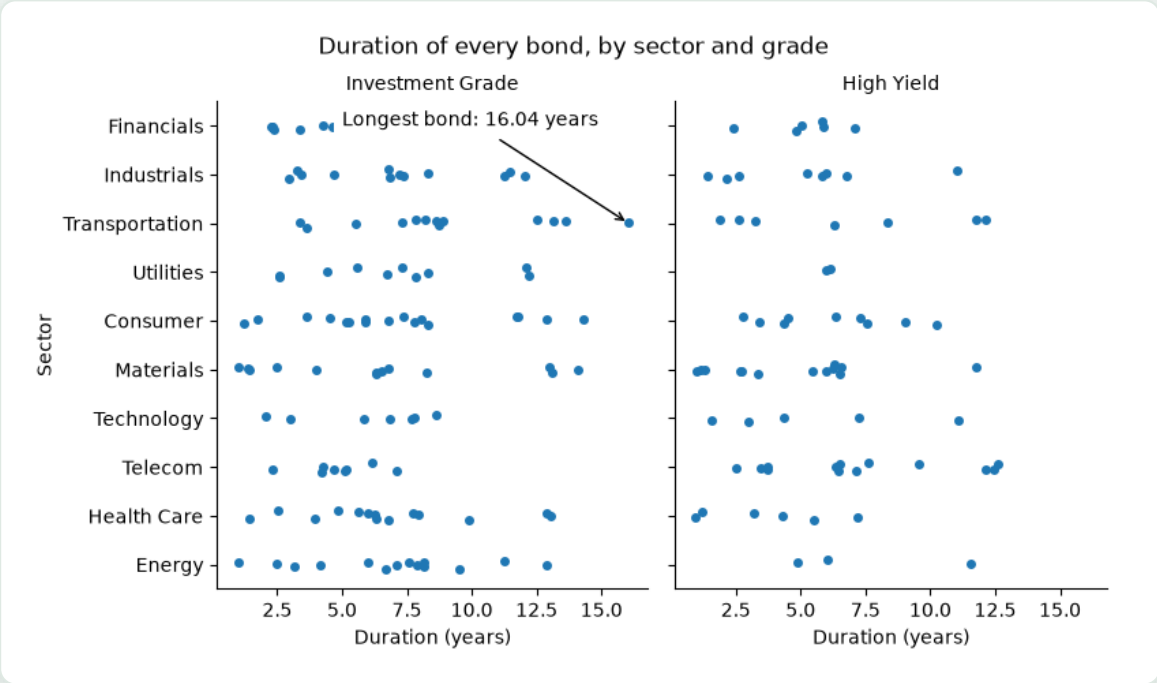
Swarm plot: every bond, no overlap




```
sns.swarmplot(data=book, x='grade',
    y='duration', ax=ax)
```

128 points on the left, 72 on the right. The top point is the longest bond, 16.04 years. Nothing is random. Excel: on a Box and Whisker chart, Format Data Series, Show inner points.

Catplot: one function, and a panel for each group




```
grid = sns.catplot(data=book, x='duration',
                   y='sector', col='grade',
                   kind='strip', height=4.5, aspect=0.9)
```

What to read off it

2

panels, one for each grade

10

sectors, on the same axis

16.042

years: the longest bond

The counts from pd.crosstab() are the points in each row. Utilities: 10 Investment Grade and 2 High Yield. Materials: 13 in each. Telecom is the one sector with more High Yield, 13 against 8. The longest bond is in Transportation, Investment Grade.

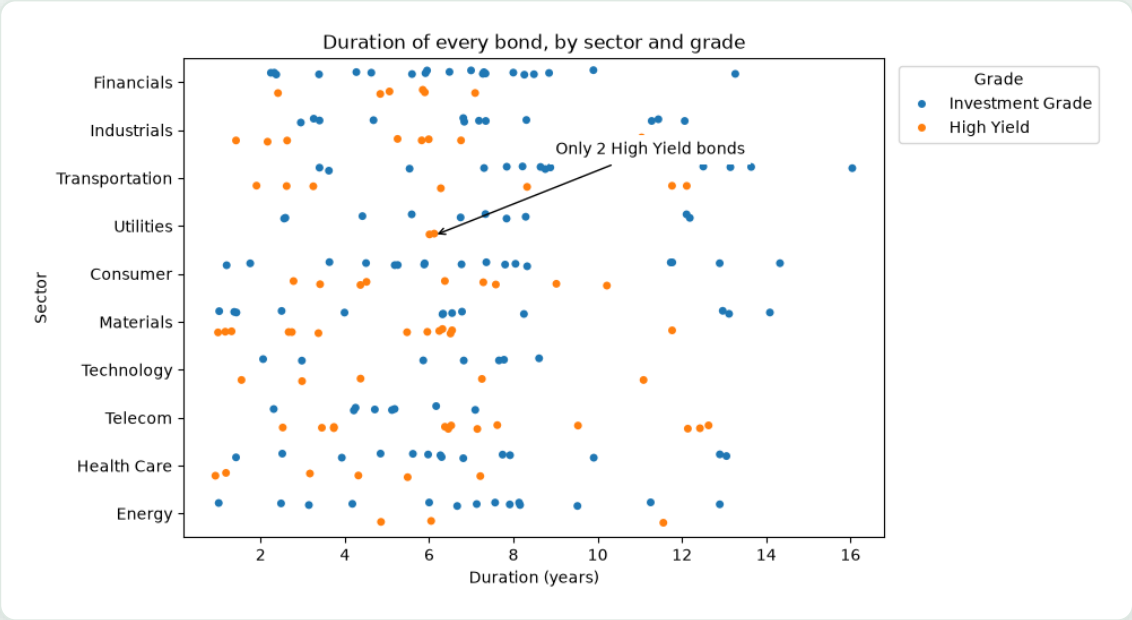
The same in Excel

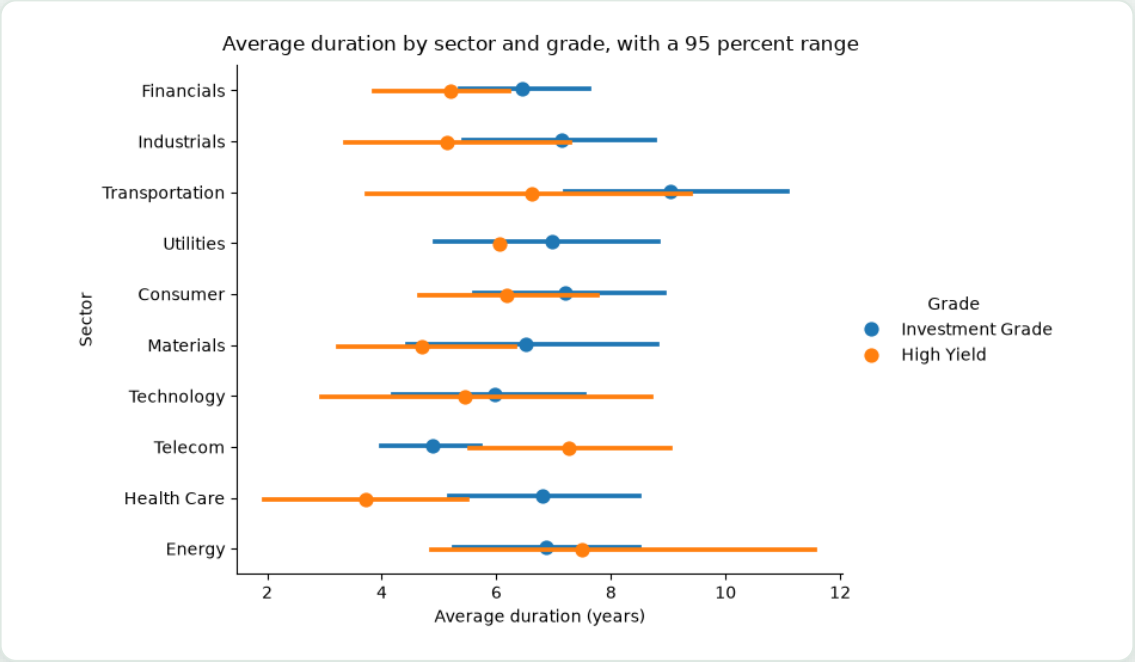
One chart per group: filter to a grade, build it, repeat

```
=COUNTIFS(E2:E201, "Utilities", V2:V201, "High Yield")
```

returns 2. Pivot table: sector in Rows, grade in Columns

Side by side, and averages with a range






```
sns.stripplot(..., hue='grade', dodge=True)
sns.catplot(..., kind='point', dodge=True)
```

A long line means the bonds of a group disagree, or there are few. Utilities High Yield is two bonds of almost the same duration, so its line is hidden.

plotly: a chart that answers to the mouse

```
fig = px.scatter(book, x='duration', y='oas', color='grade',
                 hover_name='issuer',
                 hover_data=['ticker', 'sector', 'rating'])
fig.show()
```

With the mouse	What happens
Point at a bond	a box shows its values
Drag a rectangle	zooms in on that area
Double-click	zooms back out
Click a legend name	hides or shows that group

```
fig = px.bar(value_table, x='sector', y='market_value',
             color='grade')   # stacked bars
fig.write_html('market_value_by_sector.html',
               include_plotlyjs='cdn')   # 9,285 bytes
```

The interactive chart appears when the cell is run. A static preview, such as GitHub's, may show nothing.

At the highest point, the box reads

In the box	Value
Heading	Feskovar Devices
Ticker	FESR
Sector	Technology
Rating	CCC+
Duration	1.546 years
OAS	1,551 bp, the widest in the file

A row of the invented file. The issuer does not exist.

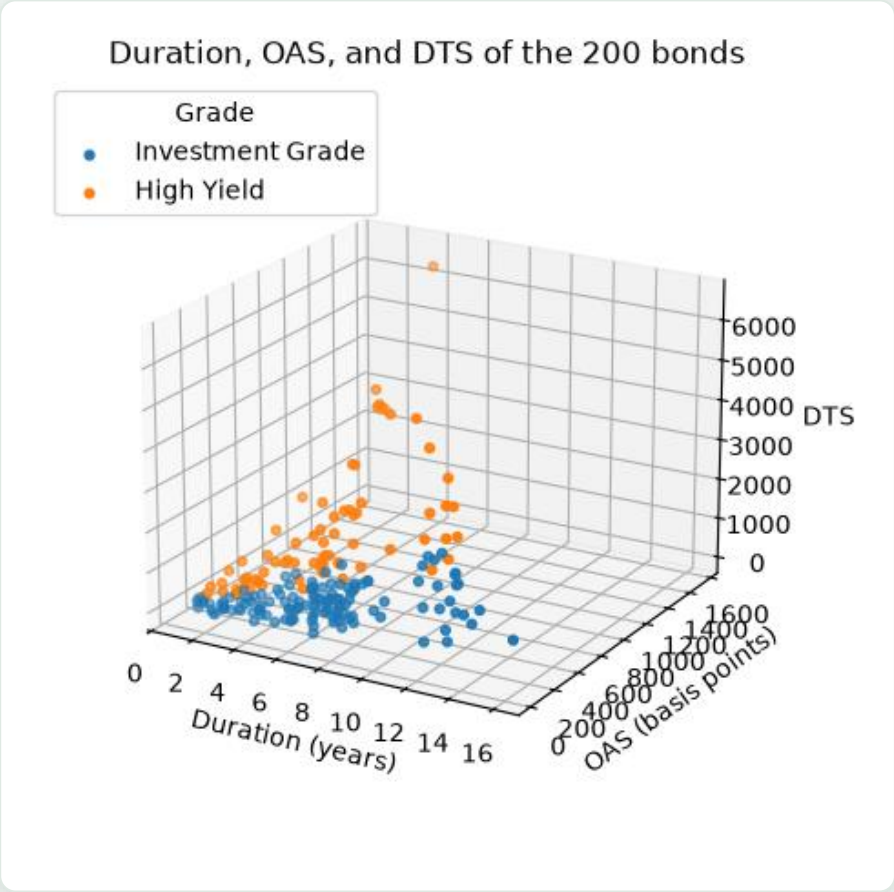
The same in Excel

Rest the pointer on a point or a bar: a tip shows the series name and the value. Stacked bars: Insert, Stacked Column, from a pivot table.

```
=SUMIFS(S2:S201, E2:E201, "Consumer",
V2:V201, "High Yield") / 1000000
```

returns 19.02: the High Yield piece of the Consumer bar

A third dimension: duration, OAS, and DTS




```
fig = px.scatter_3d(book, x='duration', y='oas', z='dts',
                    color='grade', hover_name='issuer')
fig.update_traces(marker={'size': 3})
```

What to read off it

200

bonds, one point each

0.05

largest gap, DTS to duration x OAS

6,525.8

the largest DTS, the highest point

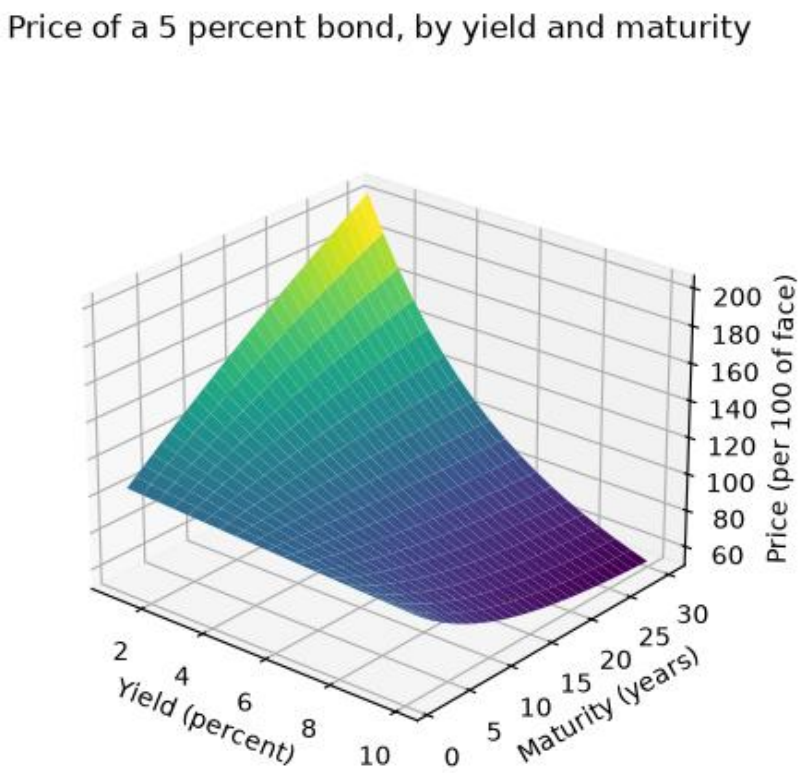
The points do not fill the space. DTS is duration times OAS, so every bond sits on the surface z = x * y. Rotate the interactive chart until you look along the sheet, and the 200 points line up.

Mouse: drag to rotate, scroll to zoom, point at a bond for its issuer

Left: matplotlib's still view. The interactive chart appears when run.

`=MAX(ABS(P2:P201 - O2:O201 * M2:M201))` returns 0.05 in Excel: rounding

A surface: price by yield and maturity




```
fig = go.Figure(go.Surface(x=yields, y=years, z=price_grid))
ax.plot_surface(yield_grid, years_grid, price_grid, cmap='viridis')
```

570

prices: 19 yields by 30 maturities

203.45

30 years, at a yield of 1 percent

52.68

30 years, at a yield of 10 percent

A 5 percent bond. Where the yield is 5 percent, the price is 100 at every maturity. At 1 year the price only runs from 95.35 to 103.97. Arithmetic on an imagined bond, not market prices.

3D helps when one quantity depends on two others and the reader can turn the chart. On a page or a slide it cannot be turned, parts hide other parts, and heights are hard to read: use a heatmap or flat panels.

Excel: a 3-D Surface chart, and PRICE, with rate and yield as decimals

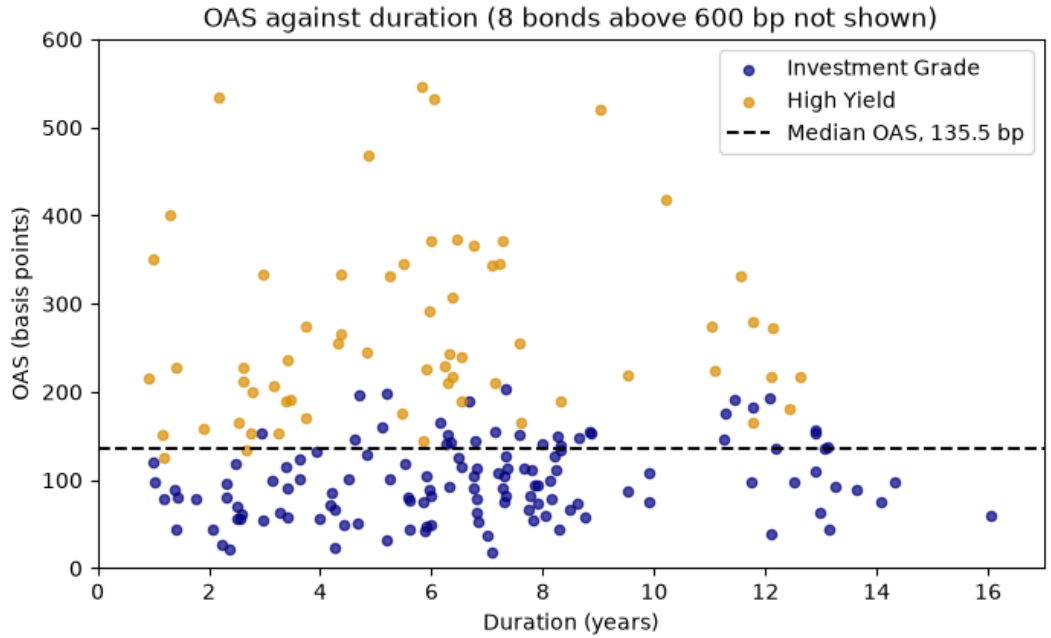
`=PRICE(DATE(2026,9,15), DATE(2056,9,15), 0.05, 0.1, 100, 2, 0)` returns 52.68

Customizing a chart: one line for each job

The job	matplotlib	Excel
Colour of a series	`color='navy'`	Format Data Series, Fill & Line
Size of the chart	`figsize=(8, 4.5)`	drag a corner, or Format Chart Area, Size
Axis limits	`ax.set_ylim(0, 600)`	Format Axis, Axis Options, Bounds: Minimum and Maximum
Thousands separators	`ax.yaxis.set_major_formatter('{x:,.0f}')`	Format Axis, Number, Format Code #,##0
Percent on an axis	`PercentFormatter(xmax=100)`	Format Axis, Number, Percentage, on values held as fractions
Value on every bar	`ax.bar_label(bars)`	Chart Elements, Data Labels
Note on one point	`ax.annotate(...)`	click the point twice, then right-click, Add Data Label
Reference line	`ax.axhline(135.5)`	a second series that holds the same value in every row
Legend position	`ax.legend(loc='upper right')`	Chart Elements, Legend, then Top, Bottom, Left, or Right
Charts side by side	`plt.subplots(1, 2)`	two charts placed next to each other on the sheet
One style for every chart	`a function such as desk_style()`	right-click a finished chart, Save as Template
Save as a picture	`fig.savefig('name.png')`	right-click the chart, Save as Picture (Microsoft 365)

In Excel the choices are made in panes and stay with that one chart, or with a template. In matplotlib they are lines of code, so the chart for next month's file comes out the same.

A chart that can leave the desk




```
ax.set_ylim(0, 600)   # the title says what was left off
ax.axhline(median_oas, color='black', linestyle='--')
# 8 bonds are above 600 bp: still in book, not drawn
```

The numbers behind this section's charts

8

bonds above 600 bp, not drawn

135.5

bp, the median OAS of the book

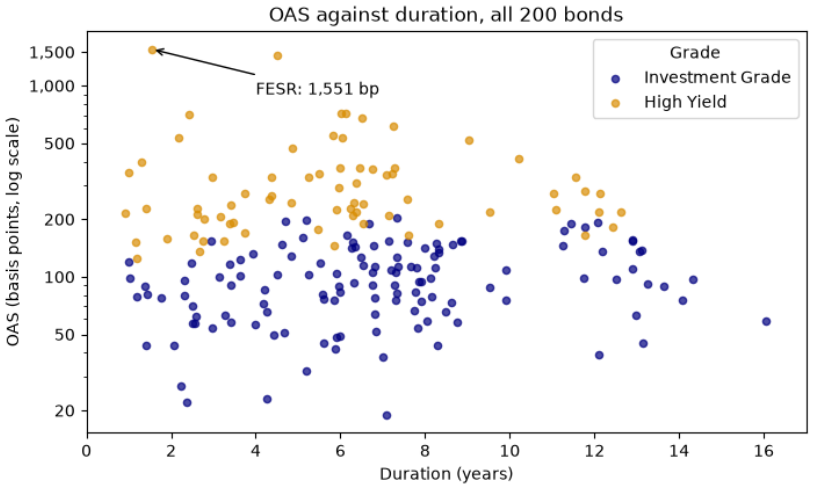
30.8%

BBB share of market value

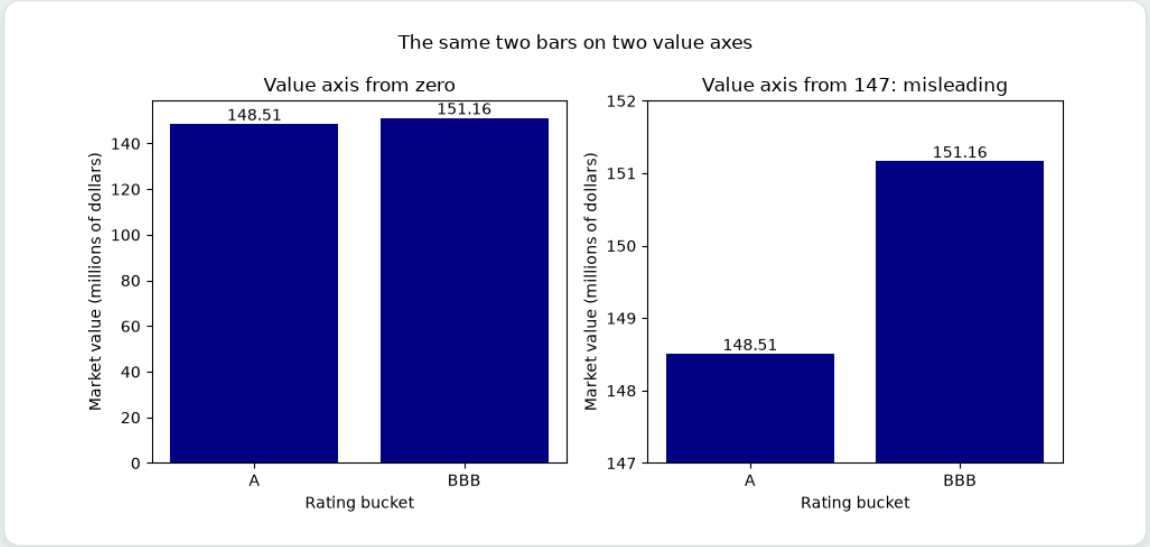
An axis is a window onto the data. When it is narrowed, say on the chart what was left off. The widest bond, ticker FESR at 1,551 bp, is a row of the invented file: ax.annotate() points an arrow at it. When panels have different scales, say so.

In Excel	Result
`=COUNTIF(M2:M201, ">600")`	8
`=MEDIAN(M2:M201)`	135.5
`=INDEX(C2:C201, MATCH(MAX(M2:M201), M2:M201, 0))`	FESR
`=SUMIF(U2:U201, "BBB", S2:S201) / SUM(S2:S201)`	30.8%

Charts that do not mislead



Log scale: all 200 bonds drawn. The axis label says so.



BBB is 1.02 times A. From 147, its bar is 2.77 times as tall.

The habit	matplotlib	Excel
A log scale, not a cut axis	`ax.set_yscale('log')`	Format Axis, Axis Options, Logarithmic scale
Bars start at zero	`the default of ax.bar()`	Format Axis, Axis Options, Bounds, Minimum set to 0
One unit on a chart	`two panels, not ax.twinx()`	two charts, not a Secondary Axis
Colours every reader can tell apart	`palette='colorblind', style='grade'`	Format Data Series, Fill & Line, Marker, Marker Options

Choosing a chart

The question	Chart	Function	Nearest in Excel
How much does y change with x?	Regression plot	`np.polyfit(), sns.regplot()`	Scatter with a Linear trendline
Two columns, and each on its own?	Joint plot	`sns.jointplot()`	a Scatter and two Histograms
What shape does each group have?	Violin plot	`sns.violinplot()`	Box and Whisker
Where is every bond of each group?	Strip or swarm plot	`sns.stripplot(), sns.swarmplot()`	Box and Whisker, Show inner points
The same chart for each group?	Panels	`sns.catplot(), plt.subplots()`	one chart per group
Which bond is that point?	Interactive chart	`px.scatter(), px.bar()`	the tip when the pointer rests on a point
One quantity against two others?	3D scatter, surface	`px.scatter_3d(), go.Surface()`	3-D Surface chart

The habits of a chart that leaves the desk

Print slope, intercept, and R squared, with units.

A fitted line describes its file. It is not a forecast.

Check a smoothed shape against the real low and high.

When an axis is narrowed, say what was left off.

When panels have different scales, say so.

One set of colours and one style, kept in a function.

Use 3D where the reader can turn it. On paper, a heatmap.

A saved chart can carry data. Treat it like the workbook.

The AI check

Every notebook ends with code written in the style of an AI assistant. It has one bug. Find it.

1

Read

Read the code before you run it.

2

Predict

Decide what it should produce.

3

Run

Run it and look at the output.

4

Check

Compare with a number you worked out yourself.

Code that runs is not the same as code that is right.

What you can do now

- Fit a straight line and state its slope, intercept, and R squared with units
- Show the shape of each group, every bond as a point, and panels by group
- Draw an interactive chart with plotly, and save it as a web page
- Draw a 3D scatter and a surface, and say when 3D is the wrong choice
- Customize a chart, keep it honest, hold the style in a function, and save it

Week 3, Day 3 Exploratory Data Analysis: sanity checks, distributions, relationships, missing values, and outliers